

# Ilja Safronovs

R&D Engineer — 3D reconstruction & computer vision · Giraffe360 · Riga, Latvia

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I work on 3D reconstruction: Gaussian splatting, and some mesh work. At Giraffe360 I build our splat pipeline from sparse panoramas and implement published methods in PyTorch and CUDA. I also work on depth models, 3D camera paths, LLM APIs and tools that help a team build faster.

## EXPERIENCE

**Giraffe360** — R&D Engineer — Riga · 2025 – present

### Fixed blurry windows and mirrors in sparse-panorama 3DGS; the default since 1 June 2026.

instagram.com/p/DZsNRB8o772

Outdoor views through the windows are fixed too. I ran 140+ controlled training runs, then 600+ experiments over a month to cut haze, floaters and flicker.

### Built our custom 3DGS pipeline from sparse panoramas, with published methods in CUDA.

I built the pipeline that turns sparse panorama views into 3D Gaussian splats. Jan 2026: influence-based densification, building on Taming 3DGS (Mallick et al., 2024). June 2026: Analytic-Splatting (Liang et al., 2024) in gsplat, with my CUDA forward and backward passes.

### Shipped two 3D video features: orbital exterior videos and the looped drone.

instagram.com/giraffe360/reel/Dd9ODBVIHhE

I led the exterior work: first results in 2 weeks, in production in 3. The looped drone was my hackathon idea (Feb 2026, special mention). I built its camera-path algorithm, and it launched in Oct 2026.

### Mentored a junior engineer on a new depth method; the whole team uses it.

Mostly his work: minima matching plus OMNI-DC to densify LiDAR depth.

### Built team tools: quality checks, an agent-driven experiment runner, an MCP server.

July 2026: ground-truth datasets and validation tools for splat quality. March 2026: an experiment runner, so every teammate can run and track training on a local GPU through an AI agent. June 2026: an MCP server (Node.js), so AI agents can read and write the team task board; colleagues use it.

### Built an embeddable Rust/wgpu splat viewer and a comparison tool.

A Rust/wgpu viewer (May 2026) that any app can embed and control, and a multi-view comparison tool (Jan 2026).

### Prototypes: mirror reflections, meshes from splats and lighter splats.

Working prototypes: correct reflections in mirrors (June 2026), mesh generation from splats, and a stage that cuts the splat count for viewers (May 2026).

### Earlier: a small LLM service that wrote tour narration.

Jul–Oct 2025: a small LLM service that wrote tour narration (OpenAI API, AWS, S3).
Giraffe360 makes cameras and software for real-estate virtual tours.

## EDUCATION

**Riga Technical University** — Professional Bachelor, Financial Engineering — 2024 – 2028 (expected)

### Maths, statistics and programming applied to finance.

Qualification: senior data analysis specialist. Maths, statistics and programming applied to finance.

Coursework: mathematical analysis and algebra, probability and statistics, econometrics, time-series analysis, stochastic analysis, Monte Carlo methods, portfolio analysis, actuarial mathematics, data structures and algorithms, databases/SQL, computer networks, programming (Java, C++).

**AI Talent Program** (Ponder) — Nov 2024 – Jan 2025

Mākslīgā intelekta talantu programma

**Selective program in C++, Python, maths and AI.**
Selective program: C++, Python, maths and AI development with industry mentors. Theory of AI: its history, neural network architectures (CNNs, transformers, GANs, VAEs), computer vision and reinforcement learning. Led to the Giraffe360 role.

**Riga State Classical Gymnasium** — Programming and Mathematics track
Rīgas Valsts klasiskā ģimnāzija

**State mathematics olympiads and the Small Mathematics University.**
Specialised track in programming and mathematics (today named "Programming + AI"). Latvian state mathematics olympiads; completed the Small Mathematics University (Mazā matemātikas universitāte) at the University of Latvia.

## PROJECTS

**load-balancing-experiments — seven experiments on DNS, queueing and load balancing.**

**flow — offline workout tracker in Dart/Flutter, in daily use.**

**monte-carlo-finance — option pricing, volatility and risk models in Python.**

**Portfolio journey — this tap-driven WebGL site in three.js.**

**Offline Graphing Calculator — 4th place of 27 in a school competition.**

crypto-quant-lab — 59 backtested trading-rule versions on 6 coins.

projects_with_NEAT — neuroevolution agents that learn Flappy Bird and Pong.

RTU_program_parser — finds extra RTU lectures that fit your schedule.

Snake + maze generator — maze generation, terrain and path-finding in Pygame.

bezier-curve-js — interactive Bezier-curve drawing tool.

sort_visualization — 16 sorting algorithms animated in C++/SFML.

## SKILLS

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| 3D & vision | Gaussian splatting, 3D reconstruction, computer vision, depth estimation, anti-aliasing, mesh generation, diffusion-based artifact fixing, PyTorch, CUDA kernels |
| Research | Paper-to-production, benchmarking and evaluation |
| Languages | Python, C++, Dart/Flutter, JavaScript, Java, SQL |
| Tools | Linux, Docker, Git/GitLab CI, AWS S3, Pygame, Selenium, pandas |
| Other | OpenAI API / LLM prompting, neuroevolution (NEAT), three.js/WebGL |
| Spoken | Latvian, English, Russian |